

BrainQuests Kids – Product Requirements Document (PRD)

1. Vision

Create a fun, safe, and educational mobile app that enhances children's cognitive abilities (memory, logic, attention, problem-solving) through quick mini-games designed for ages 4–12.

2. Target Users

- Children aged 4–12
 - Parents seeking educational content
 - Schools and nurseries looking for simple brain-training tools
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3. Core Features

3.1 Mini Games

1. **Memory Cards** – Match pairs.
2. **Find the Difference** – Spot 2–5 differences.
3. **Shape Matcher** – Drag the correct shape.
4. **Pattern Logic** – Complete the sequence.
5. **Quick Math Kids** – Simple addition/subtraction.
6. **Color Memory (Simon Style)** – Repeat the color order.

Each game includes: - 10 Levels - Stars and points - Reward after each level

3.2 Progress System

- Stars 
- Badges 
- Progress bar
- Achievement board

3.3 Rewards

- Daily Gift
- Sticker Album
- Rewards after every 3 completed levels

3.4 Ads System

- Banner ads only (AdMob)
- No pop-ups
- No forced video ads
- Premium version removes ads and unlocks extra features

3.5 Parent Mode (Future Update)

- Playtime tracking
 - Game difficulty suggestions
 - Favorite game analytics
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4. UX/UI Requirements

- Playful colors (Blue, Yellow, Green)
 - Large buttons
 - Smooth animations
 - Simple text
 - Cute sound effects
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5. Technical Requirements

- **Flutter** framework
 - Local storage for progress
 - Modular mini-game structure
 - Offline support
 - Firebase Analytics + Crashlytics
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6. Monetization

Freemium Model

- **Free:** Core games
 - **Premium (1.99\$):**
 - Remove ads
 - Unlock extra games
 - Full sticker album
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7. KPI Metrics

- Daily Active Users (DAU)
 - Monthly Active Users (MAU)
 - Session length
 - Retention (Day 1 / Day 7 / Day 30)
 - Level completion rates
 - Premium conversion
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8. Roadmap Timeline

Week 1

- PRD
- App name + logo
- UI/UX draft
- App structure setup

Week 2

- Build core mini-games
- Add progress + rewards
- Integrate AdMob

Week 3

- Testing + refinement
 - UI polish
 - Play Store listing and screenshots
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MiniGenius – Branding & Logo Concept

1. Logo Concept

- Cartoon brain icon
- Friendly smile
- Rounded edges
- Flat modern vector style
- Child-friendly colors (Blue + Yellow)
- Simple details for a clear 512×512 app icon

2. Brand Colors (Initial)

- Blue: #4DB7FF
- Yellow: #FFE45C
- Mint Green: #5ADBB5
- White: #FFFFFF
- Dark Text: #222222

3. AI Logo Prompt (Primary)

Create a cute and modern kids mobile app logo for “MiniGenius”. Design a simple cartoon brain with a friendly smile, rounded edges, and clean outlines. Use bright colors (sky blue and yellow), soft shadows, and a flat vector style. Icon should be square, centered, high-contrast, minimal details, and suitable as a mobile app icon (512x512). Make it playful, friendly, and visually appealing to children.

4. AI Logo Prompt (Alternative Variations)

Variation 1 – Character Style

Cute cartoon brain character mascot for kids app “MiniGenius”, big eyes, friendly smile, vibrant colors, soft gradients, rounded vector design, app icon style.

Variation 2 – Minimal Flat Style

Minimal flat brain icon with smile for kids app “MiniGenius”, blue and yellow colors, simple shapes, very clean, child-friendly, vector logo.

MiniGenius – UI Style Guide

1. Typography

- **Primary Title Font:** Rounded, playful sans-serif (e.g., Fredoka / Nunito)
- **Body Font:** Clean sans-serif (e.g., Poppins / Quicksand)
- **Font Characteristics:**
 - Rounded corners

- High readability for children
- Medium-weight for buttons and labels

2. Color Usage

- **Primary Blue (#4DB7FF)**: Headers, main buttons, highlights
- **Yellow (#FFE45C)**: Accents, stars, reward screens
- **Mint Green (#5ADBB5)**: Progress bars, success states
- **White (#FFFFFF)**: Backgrounds
- **Dark Text (#222222)**: Main readable text

3. Buttons

- Rounded corners (12–20px)
- High-contrast icons
- Inner shadow subtle for depth
- Hover/Pressed State:
- Slight scale animation (1.05x)
- Darker shade of the main color

4. Iconography Style

- Flat vector
 - Thick rounded outlines
 - Minimal details
 - Bright, kid-friendly shapes
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App Icon Guidelines

1. Icon Shape

- Square with rounded corners
- 10–15% padding around the brain graphic
- No text on the icon

2. Composition

- Cartoon brain centered
- Smile visible even at 48×48 resolution
- Background in gradient Blue → Mint or solid Blue

3. Avoid

- Too many details

- Text or small patterns
 - Sharp or angular shapes
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Home Screen – Wireframe Description

Layout

- **Top:** MiniGenius logo (small version)
- **Center:** 6 Game Buttons arranged 2×3 grid
- Each game icon is round and colorful
- Under each icon: tiny label (Memory, Shapes, etc.)
- **Bottom Bar:**
- Home | Stickers | Achievements | Settings
- **Background:** Soft gradient Blue

Micro-Animations

- Hover bounce on game icons
 - Stars floating in background lightly
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Game Select Screen – Wireframe Description

Structure

- **Title:** Game Name (e.g., Memory Cards)
 - **Levels Grid:** 10 circles
 - Locked levels → grey with lock icon
 - Unlocked → colored with stars
 - **Start Button:** Big, rounded, yellow
 - **Back Button:** Simple arrow
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Text Logo (Wordmark)

Concept

- Rounded, playful typography
- “Mini” in Blue
- “Genius” in Yellow
- Small smiling brain replacing the dot on the “i” in Mini

Usage

- Splash screen
 - Website header
 - App Store screenshots
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Sticker Pack – Visual Direction

Sticker Themes

- Cute Brain Characters
- Happy Brain
- Thinking Brain
- Winner Brain
- Achievement Stickers
- “Level Up!”
- “Super Smart!”
- “Mini Genius!”
- Reward Items
- Stars ★
- Trophies 🏆
- Colorful badges ★

Style

- Thick outline
- Soft shading
- Bright pastel colors
- Suitable for reward animations

Usage

- Reward screens
 - Daily gift
 - Album collection
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